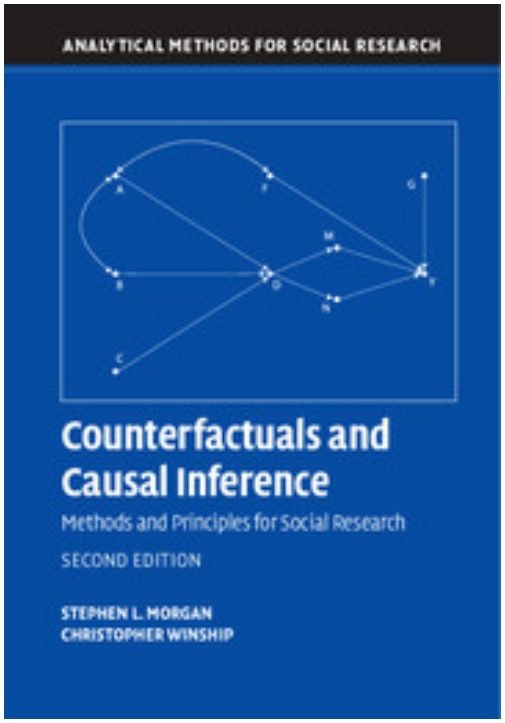
Counterfactuals and Causal Inference

Methods and Principles for Social Research



Chapter 5

Outline

1. Origins of and Motivations for Matching
2. Matching as Conditioning via Stratification
3. Matching as Weighting
4. Matching as a Data Analysis Algorithm
5. Remaining Practical Issues in Matching Analysis

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A classic technique, brought back

Four eras:

1. Something simpler than regression
2. A method of sampling for the analysis of matched pairs
3. A type of flexible nonparametric regression for the estimation of treatment effects
4. A data preprocessor, after which parametric modeling is undertaken

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"Perfect" stratification (our language for now ...)

- Suppose that those who take the treatment and those who do not are very much unlike each other, and yet the ways in which they differ are captured exhaustively by a set of observed treatment assignment/selection variables S.
- Knowledge and observation of S allow for a "perfect stratification"of the data.
- By "perfect,"we mean precisely that individuals within groups defined by values on the variables in S are entirely indistinguishable from each other in all ways except for
  1. observed treatment status
  2. differences in the potential outcomes that are independent of treatment status.

Identification assumptions

Under such a perfect stratification of the data, we can assert:

$$\text{Assumption 1-S:} \quad E[Y^1|D=1,S] = E[Y^1|D=0,S], \quad (5.1)$$
$$\text{Assumption 2-S:} \quad E[Y^0|D=1,S] = E[Y^0|D=0,S]. \quad (5.2)$$

These assumptions would suffice to enable consistent and unbiased estimation of the average treatment effect (ATE) because the treatment can be considered randomly assigned within groups defined by values on the variables in S.

If Assumption 1-S is valid, then

$$\begin{aligned} E[\delta|D=0,S] &= E[Y^1 - Y^0|D=0,S] \\ &= E[Y^1|D=0,S] - E[Y^0|D=0,S] \\ &= E[Y^1|D=1,S] - E[Y^0|D=0,S] \\ &= E[Y|D=1,S] - E[Y|D=0,S]. \end{aligned} \tag{5.3}$$

If Assumption 2-S is valid, then

$$\begin{aligned} E[\delta|D=1,S] &= E[Y^1 - Y^0|D=1,S] \\ &= E[Y^1|D=1,S] - E[Y^0|D=1,S] \\ &= E[Y^1|D=1,S] - E[Y^0|D=0,S] \\ &= E[Y|D=1,S] - E[Y|D=0,S]. \end{aligned} \tag{5.4}$$

Estimation: Within strata, then weighted across strata

- If a stratifying variable S completely accounts for all systematic differences between those who take the treatment and those who do not, then conditional-on-S estimators yield consistent and unbiased estimates of the average treatment effect conditional on a particular value s of S:

$$\{E_N[y_i|d_i = 1, s_i = s] - E_N[y_i|d_i = 0, s_i = s]\} \xrightarrow{p} E[Y^1 - Y^0|S = s] = E[\delta|S = s]. \tag{5.5}$$

- Weighted sums of these stratified estimates can then be taken, such as for the unconditional ATE:

$$\sum_s \{E_N[y_i|d_i = 1, s_i = s] - E_N[y_i|d_i = 0, s_i = s]\} \Pr_N[s_i = s] \xrightarrow{p} E[\delta] \tag{6}$$

Matching Demonstration 1

Table 5.1 The Joint Probability Distribution and Conditional Population Expectations for Matching Demonstration 1

	Joint probability distribution of S and D		
	$D = 0$	$D = 1$	
$S = 1$	$\Pr[S = 1, D = 0] = .36$	$\Pr[S = 1, D = 1] = .08$	$\Pr[S = 1] = .44$
$S = 2$	$\Pr[S = 2, D = 0] = .12$	$\Pr[S = 2, D = 1] = .12$	$\Pr[S = 2] = .24$
$S = 3$	$\Pr[S = 3, D = 0] = .12$	$\Pr[S = 3, D = 1] = .2$	$\Pr[S = 3] = .32$
	$\Pr[D = 0] = .6$	$\Pr[D = 1] = .4$	

	Potential outcomes		
	Under the control state	Under the treatment state	
$S = 1$	$E[Y^0 \mid S = 1] = 2$	$E[Y^1 \mid S = 1] = 4$	$E[Y^1 - Y^0 \mid S = 1] = 2$
$S = 2$	$E[Y^0 \mid S = 2] = 6$	$E[Y^1 \mid S = 2] = 8$	$E[Y^1 - Y^0 \mid S = 2] = 2$
$S = 3$	$E[Y^0 \mid S = 3] = 10$	$E[Y^1 \mid S = 3] = 14$	$E[Y^1 - Y^0 \mid S = 3] = 4$
	$E[Y^0 \mid D = 0] = \frac{.36}{.6}(2) + \frac{.12}{.6}(6) + \frac{.12}{.6}(10) = 4.4$	$E[Y^1 \mid D = 1] = \frac{.08}{.4}(4) + \frac{.12}{.4}(8) + \frac{.2}{.4}(14) = 10.2$	

Table 5.2 Estimated Conditional Expectations and Probabilities for Matching Demonstration 1

	Estimated mean observed outcome conditional on s_i and d_i	
	Control group	Treatment group
$s_i = 1$	$E_N[y_i \mid s_i = 1, d_i = 0] = 2$	$E_N[y_i \mid s_i = 1, d_i = 1] = 4$
$s_i = 2$	$E_N[y_i \mid s_i = 2, d_i = 0] = 6$	$E_N[y_i \mid s_i = 2, d_i = 1] = 8$
$s_i = 3$	$E_N[y_i \mid s_i = 3, d_i = 0] = 10$	$E_N[y_i \mid s_i = 3, d_i = 1] = 14$
	Estimated probability of S conditional on D	
$s_i = 1$	$\Pr_N[s_i = 1 \mid d_i = 0] = .6$	$\Pr_N[s_i = 1 \mid d_i = 1] = .2$
$s_i = 2$	$\Pr_N[s_i = 2 \mid d_i = 0] = .2$	$\Pr_N[s_i = 2 \mid d_i = 1] = .3$
$s_i = 3$	$\Pr_N[s_i = 3 \mid d_i = 0] = .2$	$\Pr_N[s_i = 3 \mid d_i = 1] = .5$

Matching Demonstration 2

Table 5.3 The Joint Probability Distribution and Conditional Population Expectations for Matching Demonstration 2

	Joint probability distribution of S and D		
	$D = 0$	$D = 1$	
$S = 1$	$\Pr[S = 1, D = 0] = .4$	$\Pr[S = 1, D = 1] = 0$	$\Pr[S = 1] = .4$
$S = 2$	$\Pr[S = 2, D = 0] = .1$	$\Pr[S = 2, D = 1] = .13$	$\Pr[S = 2] = .23$
$S = 3$	$\Pr[S = 3, D = 0] = .1$	$\Pr[S = 3, D = 1] = .27$	$\Pr[S = 3] = .37$
	$\Pr[D = 0] = .6$	$\Pr[D = 1] = .4$	

	Potential outcomes		
	Under the control state	Under the treatment state	
$S = 1$	$E[Y^0 \mid S = 1] = 2$		
$S = 2$	$E[Y^0 \mid S = 2] = 6$	$E[Y^1 \mid S = 2] = 8$	$E[Y^1 - Y^0 \mid S = 2] = 2$
$S = 3$	$E[Y^0 \mid S = 3] = 10$	$E[Y^1 \mid S = 3] = 14$	$E[Y^1 - Y^0 \mid S = 3] = 4$
	$E[Y^0 \mid D = 0]$ $= \frac{.4}{.6}(2) + \frac{.1}{.6}(6) + \frac{.1}{.6}(10)$ $= 4$	$E[Y^1 \mid D = 1]$ $= \frac{.13}{.4}(8) + \frac{.27}{.4}(14)$ $= 12.05$	

Table 5.4 Estimated Conditional Expectations and Probabilities for Matching Demonstration 2

	Estimated mean observed outcome conditional on s_i and d_i	
	Control group	Treatment group
$s_i = 1$	$E_N[y_i \mid s_i = 1, d_i = 0] = 2$	
$s_i = 2$	$E_N[y_i \mid s_i = 2, d_i = 0] = 6$	$E_N[y_i \mid s_i = 2, d_i = 1] = 8$
$s_i = 3$	$E_N[y_i \mid s_i = 3, d_i = 0] = 10$	$E_N[y_i \mid s_i = 3, d_i = 1] = 14$
	Estimated probability of S conditional on D	
$s_i = 1$	$\Pr_N[s_i = 1 \mid d_i = 0] = .667$	$\Pr_N[s_i = 1 \mid d_i = 1] = 0$
$s_i = 2$	$\Pr_N[s_i = 2 \mid d_i = 0] = .167$	$\Pr_N[s_i = 2 \mid d_i = 1] = .325$
$s_i = 3$	$\Pr_N[s_i = 3 \mid d_i = 0] = .167$	$\Pr_N[s_i = 3 \mid d_i = 1] = .675$

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Propensity scores

- Within a perfect stratification, the true propensity score is nothing other than the within-stratum probability of receiving the treatment, or $\Pr[D = 1|S]$.
- For the hypothetical example in Matching Demonstration 1, the propensity scores are

$$\Pr[D = 1|S = 1] = \frac{.08}{.44} = .182,$$
$$\Pr[D = 1|S = 2] = \frac{.12}{.24} = .5,$$
$$\Pr[D = 1|S = 3] = \frac{.2}{.32} = .625.$$

Weighting with estimated propensity scores to address sparseness

- Suppose again that a perfect stratification of the data exists and is known. In particular, Assumptions 1-S and 2-S in Equations (5.1) and (5.2) are valid, and S is observed.
- But, suppose now that
  1. there are multiple variables in S,
  2. some of the variables in S take on many values,
  3. the true propensity score is greater than 0 and less than 1 for every stratum defined by S.
- In this scenario, there may be many strata in the available data from a finite sample in which no treatment and/or no control cases are observed, even though the true propensity score is between 0 and 1 for every stratum in the population (i.e., every population-level stratum includes individuals in both the treatment and control states).

- If one has observed the variables in S, then the propensity score can be estimated using standard methods.
  - For example, one can estimate the propensity score, assuming a logistic distribution,

$$\Pr[D = 1|S] = \frac{\exp(S\phi)}{1 + \exp(S\phi)}, \tag{5.7}$$

    and invoke maximum likelihood to estimate a vector of coefficients $\hat{\phi}$.
  - One can then stratify on the index of the estimated propensity score, $e(s_i) = s_i\hat{\phi}$, or appropriately weight the data as we will now see ...

Matching Demonstration 3

Our first Monte Carlo simulation:

- A very general and useful technique for considering the strengths and weaknesses of alternative estimators
- In this case, 50,000 datasets of 10,000 respondents each, spread evenly across 10,000 strata defined by two variables A and B
- Because of the sample size, the large number of strata, and the propensity score, we have rampant sparseness in each simulated dataset
- The sparseness is a function in the propensity score ... and that is the key.

The stipulated propensity score

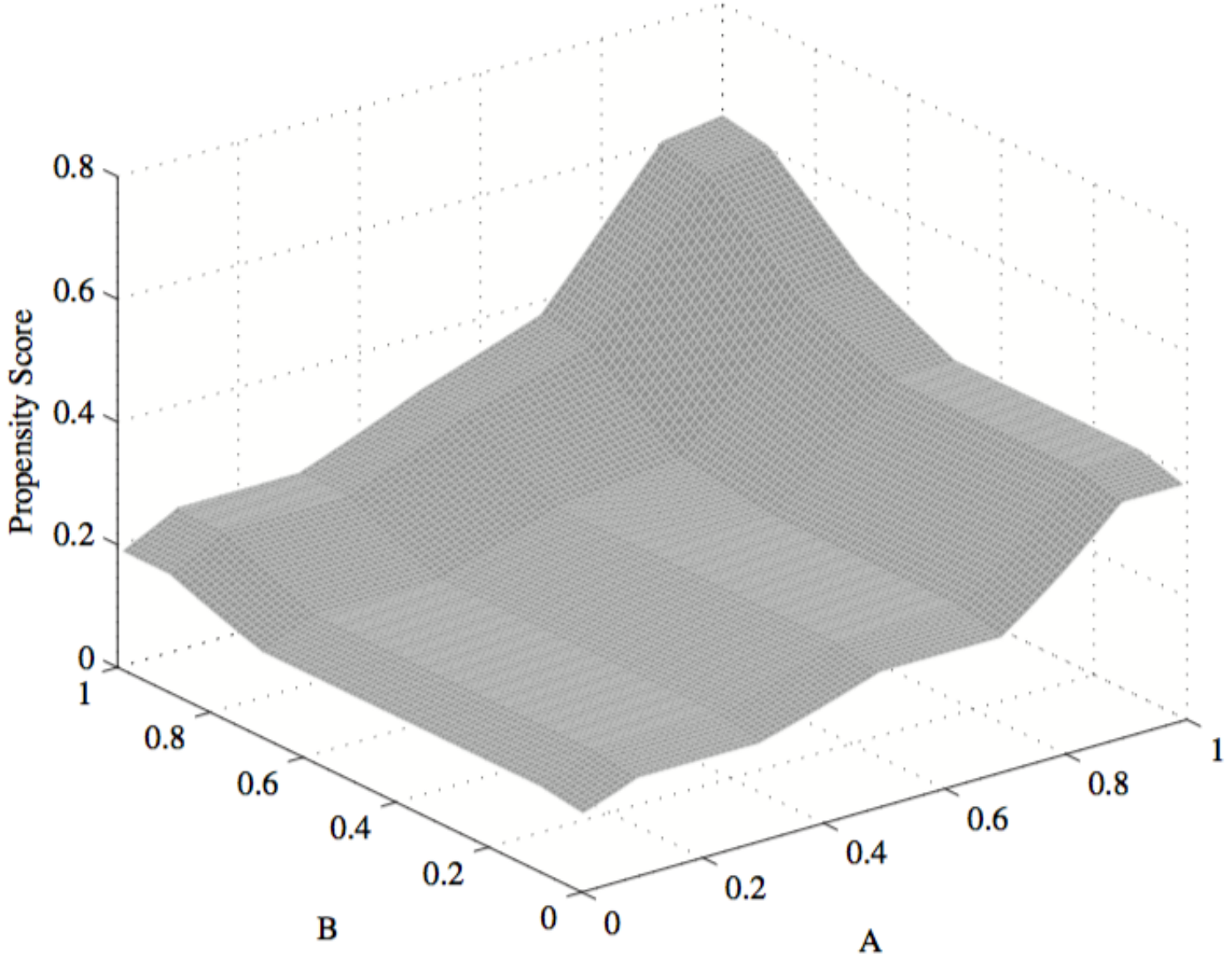

Figure 5.1 The propensity score specification for Matching Demonstration 3.

To define treatment effects for each dataset, two potential outcomes were defined as linear functions of individual values for A and B:

$$y_i^0 = 100 + 3a_i + 2b_i + \upsilon_i^0, \tag{5.8}$$

$$y_i^1 = 102 + 6a_i + 4b_i + \upsilon_i^1, \tag{5.9}$$

where both υ_i^0 and υ_i^1 are independent random draws from a normal distribution with expectation 0 and standard deviation of 5.

Look only at the top row (for now)

Table 5.5 Monte Carlo Means and Standard Deviations of True and Estimated Treatment Effects for Matching Demonstration 3

	ATE	ATT	ATC
True treatment effects	4.525	4.892	4.395
	(.071)	(.139)	(.083)
Propensity-score-based weighting estimators:			
Misspecified propensity score estimates	4.456	4.913	4.293
	(.122)	(.119)	(.128)
Perfectly specified propensity score estimates	4.526	4.892	4.396
	(.120)	(.127)	(.125)
True propensity scores	4.527	4.892	4.396
	(.127)	(.127)	(.132)

Propensity score estimation

For each of the 50,000 datasets, we us a logit model with the "wrong" specification, and we then calculate the estimated propensity score for each individual,

$$\hat{p}_i = \frac{\exp(\hat{\alpha} + \hat{\phi}_A a_i + \hat{\phi}_B b_i)}{1 + \exp(\hat{\alpha} + \hat{\phi}_A a_i + \hat{\phi}_B b_i)}, \tag{5.10}$$

along with the estimated odds of the propensity of being assigned to the treatment

$$\hat{r}_i = \frac{\hat{p}_i}{1 - \hat{p}_i}, \tag{5.11}$$

where $\hat{p}_i$ is as constructed in Equation (5.10).

Treatment effect estimation

To estimate the ATT, we then implemented a weighting estimator by calculating the average outcome for the treated and subtracting from this average outcome a counterfactual average outcome using weighted data on those from the control group:

$$\hat{\delta}_{\text{ATT,weight}} \equiv \left(\frac{1}{n^1} \sum_{i:d_i=1} y_i \right) - \left(\frac{\sum_{i:d_i=0} \hat{r}_i y_i}{\sum_{i:d_i=0} \hat{r}_i} \right), \tag{5.12}$$

where n^1 is the number of individuals in the treatment group and $\hat{r}_i$ is the estimated odds for each individual i of being in the treatment group instead of in the control group, as constructed in Equations (5.10) and (5.11).

To estimate the ATC, we then implemented a weighting estimator that is the mirror image of the one in Equation (5.12):

$$\hat{\delta}_{\text{ATC,weight}} \equiv \left(\frac{\sum_{i:d_i=1} y_i/\hat{r}_i}{\sum_{i:d_i=1} 1/\hat{r}_i} \right) - \left(\frac{1}{n^0} \sum_{i:d_i=0} y_i \right), \tag{5.13}$$

where n^0 is the number of individuals in the control group. Finally, the corresponding estimator of the unconditional ATE is

$$\begin{aligned} \hat{\delta}_{\text{ATE,weight}} \equiv & \left(\frac{1}{n} \sum_i d_i \right) \left(\hat{\delta}_{\text{ATT,weight}} \right) \\ & + \left[\left(1 - \frac{1}{n} \sum_i d_i \right) \right] \left(\hat{\delta}_{\text{ATC,weight}} \right), \end{aligned} \tag{5.14}$$

where $\hat{\delta}_{\text{ATT,weight}}$ and $\hat{\delta}_{\text{ATC,weight}}$ are as defined in Equations (5.12) and (5.13), respectively.

Monte Carlo results

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Matching estimators of the ATT

All matching estimators of the ATT can be expressed as some variation of

$$\hat{\delta}_{\text{ATT,match}} = \frac{1}{n^1} \sum_i \left[(y_i | d_i = 1) - \sum_j \omega_{i,j} (y_j | d_j = 0) \right], \qquad (5.15)$$

where n^1 is the number of treatment cases, i is the index over treatment cases, j is the index over control cases, and $\omega_{i,j}$ represents a set of scaled weights that measure the distance between each control case and the target treatment case.

Matching estimators of the ATC

For the ATC, an opposite expression is available, with alternative weights $\omega_{j,i}$ instead attached to the control cases

$$\hat{\delta}_{\text{ATC,match}} = \frac{1}{n^0} \sum_j \left[(y_j | d_j = 0) - \sum_i \omega_{j,i} (y_i | d_i = 1) \right], \qquad (5.16)$$

and where n^0 is the number of control cases.

Basic variants of matching algorithms

- Exact matching
- Nearest-neighbor, caliper, and radius matching
- Interval matching
- Kernel matching

Recent matching routines that seek optimal balance

- The recent methodological literature on matching assumes considers an omnibus performance criterion that motivates most recent matching routines: the capacity of matching estimators to balance the variables that have been matched on.
- These techniques use balance as an explicit criterion for the specification of the matching algorithm, iterating toward a final match
- We need to learn a bit more about how to think about balance ...

When will within-sample imbalance remain?

- Consider a case where we have many variables in the perfect stratification set S, where all of these variables are observed, but where we have a finite sample afflicted by rampant sparseness.
- Let $\Pr_N[s_i]$ represent the observed joint distribution across the realized values s of all variables in S for a particular sample of size N.
- For a dataset with substantial sparseness, the joint distribution $\Pr_N[s_i]$ will not in general be equal to the population distribution of S, denoted $\Pr[S]$. Generic sampling variation will lead to over-representation and under-representation of most combinations of values on the variables in S, and many of the rarest combinations in the population will not be present in the sample.

- To now consider within-sample balance with respect to a treatment effect parameter, we must consider the observed joint distribution of S conditional on membership in the observed treatment and control groups, $\Pr_N[s_i|d_i = 1]$ and $\Pr_N[s_i|d_i = 0]$, respectively.
- For all examples considered so far in this chapter, the observed data are imbalanced with respect to treatment effect estimates for D because

$$\Pr_N[s_i|d_i = 1] \neq \Pr_N[s_i|d_i = 0]. \tag{5.17}$$

In words, the joint distributions for the observed versions of the variables in the perfect stratification set S are not the same in the treatment and control groups.

- The immediate goal of all matching estimators is to transform the data in such a way that they can be analyzed as if they are balanced with respect to the treatment effect of interest, which will be the case if

$$\Pr_N[s_i|d_i = 1] = \Pr_N[s_i|d_i = 0] \tag{5.18}$$

  in the matched dataset.
- The foundational propensity-score literature claims only that the observed data will be balanced after propensity-score matching on average over repeated sampling from the same population. For any single finite sample, it is likely that the observed data will remain unbalanced to some degree.
- But ... it is still the case that you want to optimize within-sample balance. (See Matching Demonstration 3, where true propensity scores had larger MSE.)

The first step for within-sample balance assessment

- Balance should first be assessed based on the the standardized difference in sample means,

$$\frac{|E_N[x_i|d_i = 1] - E_N[x_i|d_i = 0]|}{\sqrt{\frac{1}{2}\mathrm{Var}_N[x_i|d_i = 1] + \frac{1}{2}\mathrm{Var}_N[x_i|d_i = 0]}}, \tag{5.19}$$

  which is calculated twice for every variable X that is matched on:

  1. before any matching is performed in order to establish the baseline level of imbalance in the original data and
  2. across the matched treatment and control cases to determine how much of the initial imbalance has been removed by matching.

The key recent ideas

- Expand balance assessment beyond standardized means and let that refined assessment guide specifications (perhaps automatically)
- Optimize the matching algorithm, conditional on whatever distance metric is chosen
- Directly coarsen the matching variables and then perform an exact match (typically on a subset of the sample)

The BIG question: Which of these matching algorithms works best? Do the newer methods outperform traditional algorithms?

Matching Demonstration 4

For the potential outcome under the control (i.e., a public school education), we generated what-if test scores as

$$
\begin{aligned}
y_i^0 = {} & 100 + 2(Asian_i) - 3(Hispanic_i) - 4(Black_i) \\
& - 5(Native\ American_i) - 1(Urban_i) + .5(Northeast_i) \\
& + .5(North\ Central_i) - .5(South_i) + .02(Number\ of\ Siblings_i) \\
& + .05(Own\ Bedroom_i) + 1(Two\ Parent\ Household_i) \\
& + 2(Socioeconomic\ Status_i) + 4(Cognitive\ Skills_i) + \upsilon_i^0,
\end{aligned}
\tag{5.20}
$$

where the values for υ_i^0 are independent random draws from a normal distribution with expectation 0 and standard deviation of 12

- The potential outcomes under the treatment can be written as

$$y_i^1 = y_i^0 + \delta_i' + \delta_i'', \tag{5.21}$$

where the values for δ_i' are independent random draws from a normal distribution with expectation 6 and standard deviation of 0.5.

- We added a second component with values for δ_i'', which is also a random draw from a normal distribution. In this case, however, we also set the expectation of the reference normal distribution to vary across individuals as

$$\begin{aligned} &0 + 1(Hispanic_i \times Northeast_i) \\ &+ .5(Hispanic_i \times North\,Central_i) \\ &+ 1.5(Black_i \times Northeast_i) \\ &+ .75(Black_i \times North\,Central_i) \\ &+ .5(Cognitive\,Skills_i) \end{aligned} \tag{5.22}$$

The probability of attending a Catholic school is specified using a logistic regression,

$$\Pr[D_i = 1|S_i] = \frac{\exp(S_i\phi)}{1 + \exp(S_i\phi)}, \tag{5.23}$$

where $S_i\phi$ is ...

$$
\begin{aligned}
&- 4.6 - .69(Asian_i) + .23(Hispanic_i) - .76(Black_i) \qquad (5.24)\\
&- .46(Native\ American_i) + 2.7(Urban_i) + 1.5(Northeast_i)\\
&+ 1.3(North\ Central_i) + .35(South_i)\\
&- .02(Number\ of\ Siblings_i) - .018(Own\ Bedroom_i)\\
&+ .31(Two\ Parent\ Household_i) + .39(Socioeconomic\ Status_i)\\
&+ .33(Cognitive\ Skills_i) - .032(Socioeconomic\ Status_i^2)\\
&- .32(Cognitive\ Skills_i^2)\\
&- .084(Socioeconomic\ Status_i \times Cognitive\ Skills_i)\\
&- .37(Two\ Parent\ Household_i \times Black_i)\\
&+ 1.6(Northeast_i \times Black_i) - .38(North\ Central_i \times Black_i)\\
&+ .72(South_i \times Black_i) + .23(Two\ Parent\ Household_i \times Hispanic)\\
&- .74(Northeast_i \times Hispanic_i) - 1.3(North\ Central_i \times Hispanic_i)\\
&- 1.3(South_i \times Hispanic_i) + .25\delta_i''
\end{aligned}
$$

The ATE is not identified because of self-selection

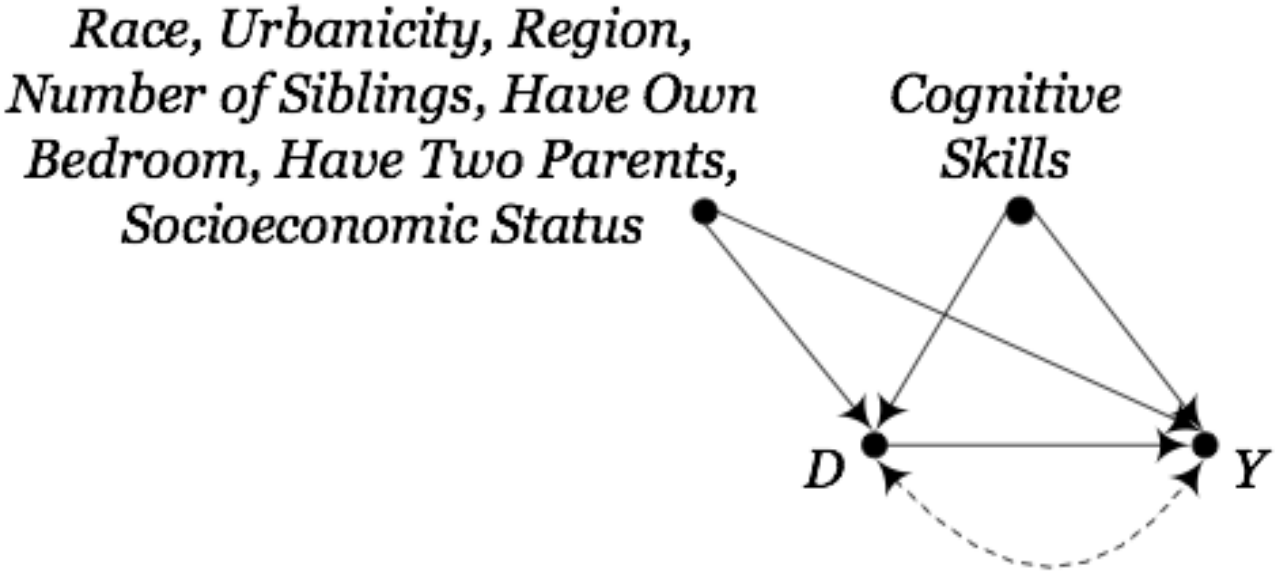

Figure 5.2 The directed graph implied by the underlying data generation model for Matching Demonstration 4.

And this is because the ATC is not identified

- Because of the way in which the generation of y_i^1 and d_i are entangled by their joint dependence on δ_i'', we cannot estimate reasonable counterfactual values for y_i^1 among those who are enrolled in public schools.
- Net of all of the other observed variables, the students most likely to benefit from Catholic schooling are in Catholic schools, and they therefore have values for $y_i = y_i^1$ that are too positive, even net of the observables, to enable effective estimation of the counterfactual values of y_i^1 for public school students.

But the ATT is identified!

- The set of variables that was used to generate y_i^0 in Equation (5.20) does not include the individual-specific shock, δ_i''.
- We therefore do not need to observe δ_i'' in order estimate reasonable counterfactual values for y_i^0 among simulated students enrolled in Catholic schools.

Methods for treatment effect estimation

- Alternative matching routines
- Two sets of treatment assignment variables: incomplete and complete
- A small Monte Carlo simulation: 1 at first, then 10
- The ATT is the target parameter

Results for a single simulated dataset (ATT=6.92)

Table 5.6 Matching Estimates of the ATT, Catholic Schooling on Achievement for One Simulated Dataset

Method	Specification of treatment assignment variables: Incomplete	Complete	Number of treatment cases retained for the estimate of the ATT
Nearest-neighbor match:			
1 without replacement (ps2)	7.37	7.03	1052
1 without replacement (MI)	7.55	7.09	1052
1 with replacement (ps2/psc)	7.77	7.17	1052
1 with replacement (MI)	7.96	7.38	1052
1 with replacement and caliper = .05 SD (ps2)	7.77	7.15	1051
1 with replacement and caliper = .05 SD (MI)	7.27	6.43	1051
5 with replacement (ps2)	7.51	6.42	1052
5 with replacement (MI)	8.06	7.29	1052
5 with replacement and caliper = .05 SD (ps2)	7.55	6.37	1051
5 with replacement and caliper = .05 SD (MI)	8.00	7.12	1051
Radius match:			
Caliper = .05 SD (ps2)	7.61	6.36	1051
Caliper = .05 SD (psc)	8.23	7.84	1051
Interval match:			
10 fixed blocks (MI)	8.71	8.71	1052
Variable blocks (ps2)	7.50	6.60	1052
Kernel match:			
Epanechnikov (ps2/psc)	7.57	6.58	1052
Gaussian (ps2/psc)	7.70	6.82	1052
Optimal match (MI-opt)	6.84	6.78	1052
Genetic match (MI-gen)	7.80	6.46	1052

Results for 10 simulated datasets

Table 5.7 Bias for Matching Estimates of the ATT, Catholic Schooling on Achievement Across 10 Simulated Datasets

	Specification of treatment assignment variables:			
	Incomplete specification		Complete specification	
Method	Min, Max	Average	Min, Max	Average
Nearest-neighbor match:				
1 without replacement (ps2)	-.37, 2.17	.79	-1.17, 0.68	-.04
1 without replacement (MI)	-.23, 2.08	.75	-1.11, 0.56	.00
1 with replacement (ps2/psc)	-.35, 2.00	.77	-1.18, 0.40	-.13
1 with replacement (MI)	-.25, 2.22	.95	-.70, 0.84	.19
1 with replacement and caliper = .05 SD (ps2)	-.32, 2.04	.78	-1.21, 0.35	-.13
1 with replacement and caliper = .05 SD (MI)	-.16, 1.89	.98	-.99, 1.16	.07
5 with replacement (ps2)	-.01, 1.77	.79	-.79, 0.83	-.04
5 with replacement (MI)	.43, 2.29	1.17	-.44, 1.34	.50
5 with replacement and caliper = .05 SD (ps2)	-.01, 1.71	.77	-.81, 0.75	-.04
5 with replacement and caliper = .05 SD (MI)	.15, 2.43	1.19	-.49, 1.18	.39
Radius match:				
Caliper = .05 SD (ps2)	-.18, 2.07	.81	-.82, 1.02	-.08
Caliper = .05 SD (psc)	.36, 2.42	1.17	.10, 2.03	.89
Interval match:				
10 fixed blocks (MI)	.84, 3.02	1.68	.84, 3.02	1.68
Variable blocks (ps2)	.03, 2.18	.88	-.85, 1.12	-.03
Kernel match:				
Epanechnikov (ps2/psc)	.08, 2.25	.89	-.76, 1.25	.01
Gaussian (ps2/psc)	.08, 2.34	1.00	-.59, 1.44	.19
Optimal match (MI-opt)	.45, 2.89	1.28	-.36, 1.53	.54
Genetic match (MI-gen)	.09, 1.66	.96	-.89, 1.55	.23
Coarsened exact match (cem)	.58, 2.66	1.28	-.43, 1.59	.35

Notes: See notes to Table 5.6 for software details.

Outline

1. Origins of and Motivations for Matching
2. Matching as Conditioning via Stratification
3. Matching as Weighting
4. Matching as a Data Analysis Algorithm
5. Remaining Practical Issues in Matching Analysis

Matching when treatment assignment is nonignorable

- You can do it, but be careful.
- Recognize that your conclusions are provisional, and consider a sensitivity analysis (to be discussed later).

Matching only on the region of common support

- Might be a good idea, but you need to assess the consequences for your conclusions of dropping cases.

The expected variance of matching estimates

- We still await the work of statisticians for a unified framework.
- The prudent advice: do not develop conclusions that depend too heavily on significance tests.

Matching estimators for many-valued causes

- Painful, and hence rare, but of course possible.
- Be careful to match appropriately.